



BITS Pilani

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CE F231 Fluid Mechanics

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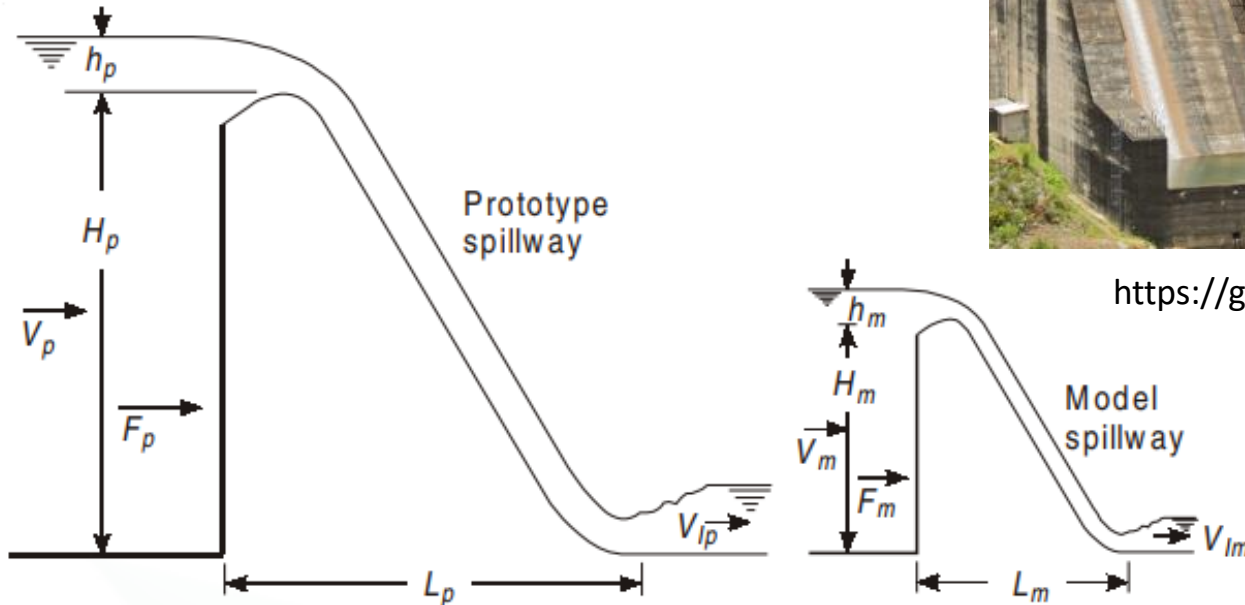
December 14, 2022



Lecture 36: Dimensional and Model Analyses

Model analysis

- Predict performance of structures before actually constructing the structures
- Prototype: Actual structure
- Model: Small-scale replica



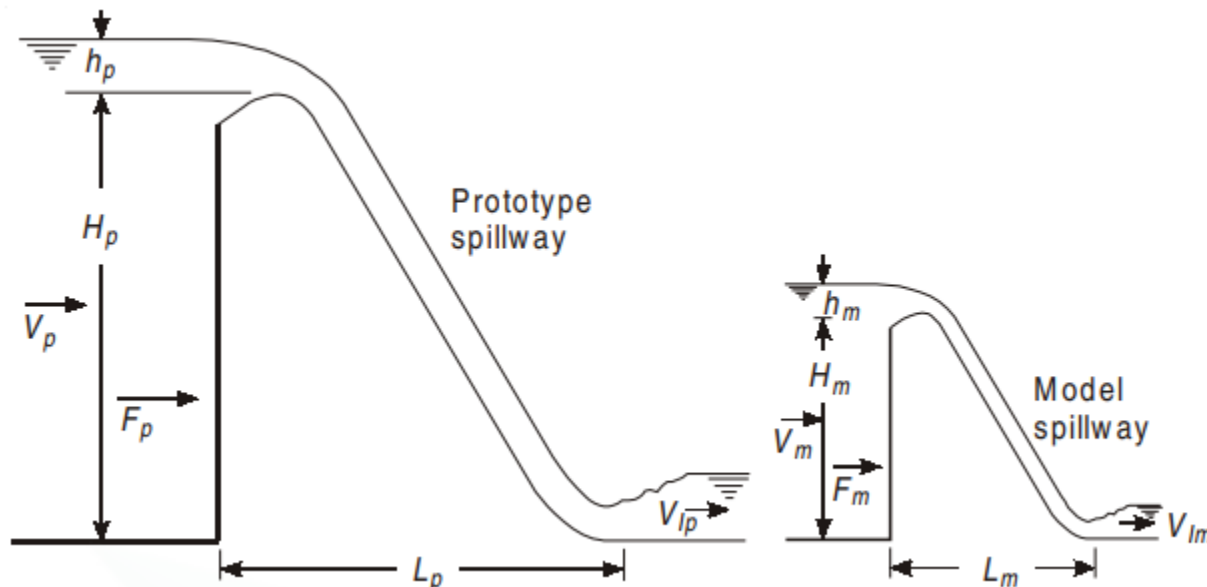
<https://graconllc.com/what-is-a-dam-spillway/>

Model analysis: benefits

- Useful for problems that cannot be solved using theoretical analysis
- Model tests are economical and convenient
- Beneficial for optimizing design
- Useful information may be obtained if complete similarity between model and prototype exists

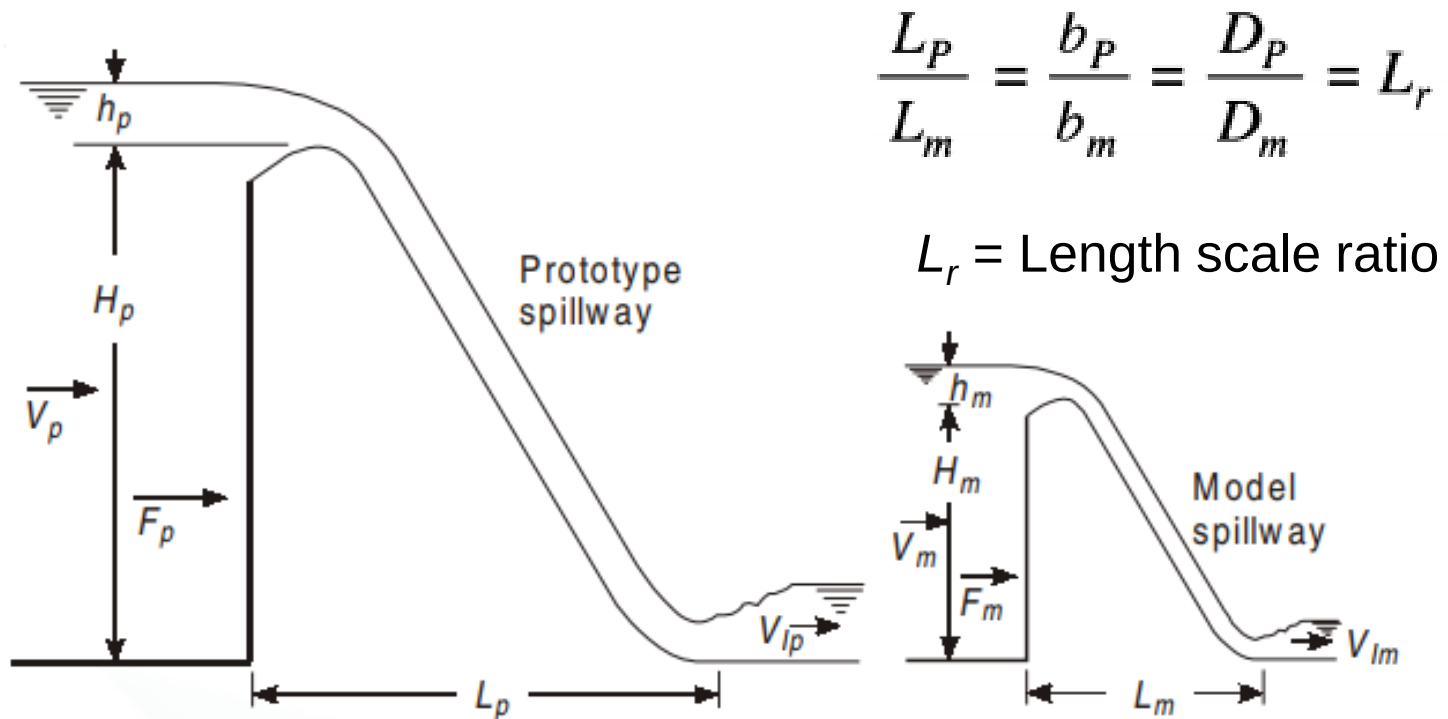
Similitude

- Similitude: similarity between model and prototype in every respect
- Three types of similarities: geometric, kinematic and dynamic



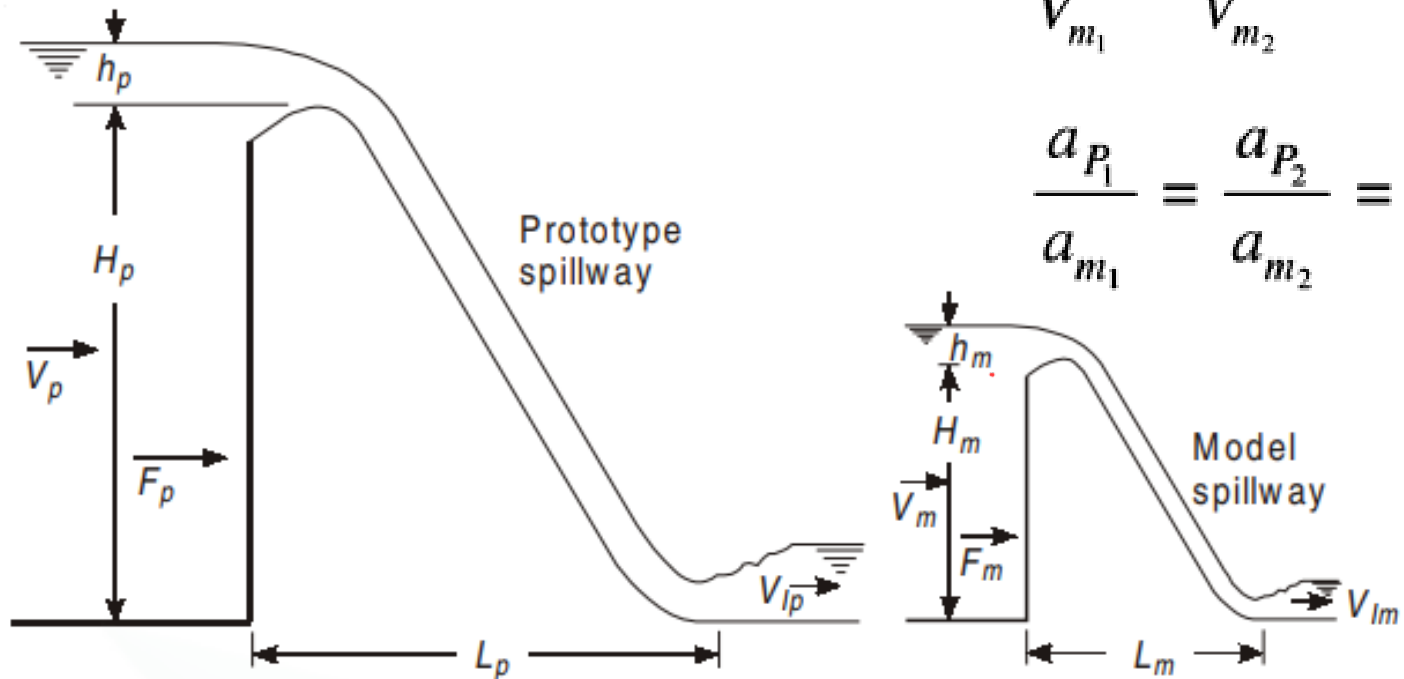
Similitude

- Geometric similarity: Ratio of all corresponding linear dimensions in model and prototype are equal



Similitude

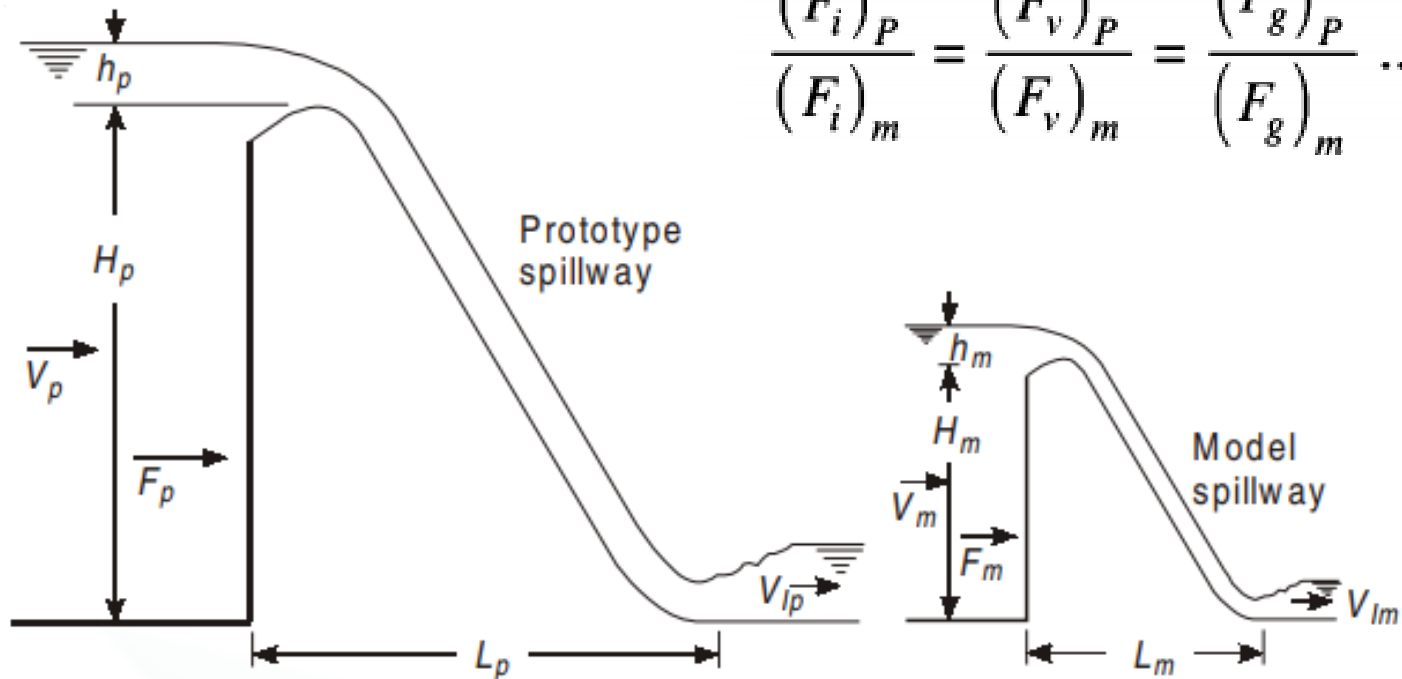
- Kinematic similarity: Ratio of velocity (and acceleration) at corresponding points in the model and prototype are same



Similitude

- Dynamic similarity: Ratio of all corresponding forces in geometrically and kinematically similar model and prototype are equal

$$\frac{(F_i)_P}{(F_i)_m} = \frac{(F_v)_P}{(F_v)_m} = \frac{(F_g)_P}{(F_g)_m} \dots = F_r,$$



Types of forces

- One or combination of following forces may be acting
 - Inertia force: Mass times acceleration forces
 - Viscous force: Shear stress due to viscosity times surface area
 - Gravity force: Mass time acceleration due to gravity
 - Pressure force: Pressure intensity times area
 - Surface tension force: Surface tension times length
 - Elastic force: Elastic stress times area



Force ratios: dimensionless numbers

- Reynold number: Inertial and viscous forces ratio $\frac{\rho VL}{\mu}$
- Froude number: Inertial and gravity forces ratio $\frac{V}{\sqrt{gL}}$
- Euler number: Inertial and pressure forces ratio $\frac{V}{\sqrt{p/\rho}}$
- Mach number: Inertial and elastic forces ratio $\frac{V}{C}$
- Weber number: Inertial and surface tension forces ratio $\frac{V}{\sqrt{\sigma/\rho L}}$

Model laws or similarity laws

- Model laws are used to transfer results obtained from models to prototypes
- Developed based on dynamic similarity b/w model and prototype
- In reality, all types of forces are not simultaneously predominant
- One force in addition to inertial force is significant
- Equal values of dimensionless parameter for model and prototype: flow pattern in model and prototype are similar

Model laws or similarity laws

- Reynold's model law (inertial and viscous): viscous flow through pipes, resistance experienced by submarines,

etc.

$$\frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_P V_P L_P}{\mu_P}$$

- Froude's model law (inertial and gravity): flow over spillway, weirs, channels, etc.

$$\frac{V_m}{\sqrt{g_m L_m}} = \frac{V_P}{\sqrt{g_P L_P}}$$

Model laws or similarity laws

- Euler's model law (inertial and pressure): turbulent flow in closed pipe where viscous and gravity forces are negligible

$$\frac{V_m}{\sqrt{p_m / \rho_m}} = \frac{V_P}{\sqrt{p_P / \rho_P}}$$

- Weber's model law (inertial and surface tension): capillary rise in narrow passages, capillary movement of water in soils

$$\frac{V_m}{\sqrt{\sigma_m / \rho_m L_m}} = \frac{V}{\sqrt{\sigma_P / \rho_P L_P}}$$

Model laws or similarity laws

- Mach's model law (inertial and elastic): Airplane through air at supersonic speed, testing of torpedo underwater etc.

$$\frac{V_m}{\sqrt{K_m / \rho_m}} = \frac{V}{\sqrt{K_P / \rho_P}}$$

Example

A model of a sub-marine of scale $\frac{1}{40}$ is tested in a wind tunnel. Find the speed of air in wind tunnel if the speed of sub-marine in sea-water is 15 m/s. Also find the ratio of the resistance between the model and its prototype. Take the values of kinematic viscosities for sea-water and air as .012 stokes and 0.016 stokes respectively. The density of sea-water and of air are given as 1030 kg/m³ and 1.24 kg/m³ respectively.

$$\frac{L_m}{L_p} = \frac{1}{40}$$

$$V_m = ?$$

$$V_p = 15 \text{ m/s.}$$

$$\left[\text{Ans. } 800 \text{ m/s, } \frac{F_m}{F_p} = 0.00214 \right]$$

$$\frac{F_m}{F_p} = ?$$

$$\nu_p = 0.012 \text{ stokes}$$

$$\nu_m = 0.016 \text{ stokes}$$

$$\rho_p = 1030 \text{ kg/m}^3$$

$$\rho_m = 1.24 \text{ kg/m}^3$$

$$\frac{\rho_m V_m L_m}{\mu_m}$$

$$= \frac{\rho_p V_p L_p}{\mu_p}$$

Example

A model of a sub-marine of scale $\frac{1}{40}$ is tested in a wind tunnel. Find the speed of air in wind tunnel if the speed of sub-marine in sea-water is 15 m/s. Also find the ratio of the resistance between the model and its prototype. Take the values of kinematic viscosities for sea-water and air as .012 stokes and 0.016 stokes respectively. The density of sea-water and of air are given as 1030 kg/m³ and 1.24 kg/m³ respectively.

$$\frac{V_m L_m}{\nu_m} = \frac{V_p L_p}{\nu_p} \Rightarrow V_m \frac{L_m}{L_p} = V_p \frac{\nu_m}{\nu_p} \quad \left[\text{Ans. } 800 \text{ m/s, } \frac{F_m}{F_p} = 0.00214 \right]$$

$$\Rightarrow V_m \times \frac{1}{40} = 15 \times \frac{0.016}{0.012}$$

$$\Rightarrow V_m = 800 \text{ m/s.}$$

$$\frac{F_m}{F_p} = \frac{\rho_m \frac{dV_m}{dt} \times L_m^2}{\rho_p \frac{dV_p}{dt} \times L_p^2}$$

$$= \frac{\rho_m V_m L_m^2}{\rho_p V_p L_p^2}$$

$$= \frac{\rho_m V_m L_m}{\rho_p V_p L_p}$$

$$= \frac{\rho_m V_m L_m}{\rho_p V_p L_p}$$

Example

A model of a sub-marine of scale $\frac{1}{40}$ is tested in a wind tunnel. Find the speed of air in wind tunnel if the speed of sub-marine in sea-water is 15 m/s. Also find the ratio of the resistance between the model and its prototype. Take the values of kinematic viscosities for sea-water and air as .012 stokes and 0.016 stokes respectively. The density of sea-water and of air are given as 1030 kg/m³ and 1.24 kg/m³ respectively.

$$\frac{F_m}{F_p} = \frac{M_m V_m L_m}{M_p V_p L_p}$$

$$\left[\text{Ans. } 800 \text{ m/s, } \frac{F_m}{F_p} = 0.00214 \right]$$

$$= \frac{\rho_m \nu_m V_m L_m}{\rho_p \nu_p V_p L_p} = \frac{1.24 \times 0.016 \times 800}{1030 \times 0.012 \times 15} \times \frac{1}{40}$$

$$= 0.00214$$

Model laws or similarity laws

➤ Reynold's model law: pipe flow, resistance experienced by submarines, etc.

➤ Froude's model law: flow over spillway, weirs, channels, etc.

$$\frac{F_p}{F_m} = \left(\frac{\mu_p}{\mu_m} \right)^2 \frac{\rho_p}{\rho_m} = \frac{v_p^2 \rho_p^2}{v_m^2 \rho_m^2} \frac{\rho_m}{\rho_p} = \left(\frac{0.012}{0.016} \right)^2 \times \frac{1030}{1.24}$$

TABLE 17.2 Scale Ratios for Models Governed by Reynolds and Froude Model Laws Scale Ratios

Description of Quantities	Reynolds Law	Froude Law
Length	L_r	L_r
Velocity	$\mu_r / L_r \rho_r$	$L_r^{1/2} g_r^{1/2}$
Time	$L_r^2 \rho_r / \mu_r$	$L_r^{1/2} / g_r^{1/2}$
Acceleration	$\mu_r^2 / \rho_r^2 L_r^3$	g_r
Discharge	$L_r \mu_r / \rho_r$	$L_r^{5/2} g_r^{1/2}$
Force	μ_r^2 / ρ_r	$\rho_r L_r^3 g_r$
Work, Energy and Torque	$\mu_r^2 L_r / \rho_r$	$\rho_r L_r^4 g_r$
Pressure intensity	$\mu_r^2 / L_r^2 \rho_r$	$\rho_r g_r L_r$
Power	$\mu_r^3 / \rho_r^2 L_r$	$\rho_r L_r^{7/2} g_r^{3/2}$

$$= 467.237$$

$$\Rightarrow \frac{F_m}{F_p} = \frac{1}{467.237}$$

$$= 0.00214$$

Example

A spillway model is to be built to a geometrically similar scale of $\frac{1}{40}$ across a flume, of 50 cm width. The prototype is 20 m high and maximum head on it is expected to be 2 m. (i) What height of model and what head on the model should be used ? (ii) If the flow over the model at a particular head is 10 litres/s, what flow per metre length of the prototype is expected ? (iii) If the negative pressure in the model is 150 mm, what is the negative pressure in the prototype ? Is it practicable ?

[Ans. (i) 0.5 m, 0.05 m, (ii) 5059.64 litres/s (iii) 6 m. Yes]

$$\frac{L_m}{L_p} = \frac{1}{40} = \frac{H_m}{20} \Rightarrow H_m = 0.5 \text{ m}$$

$$\frac{L_m}{L_p} = \frac{h_m}{h_p} = \frac{1}{40} \Rightarrow \frac{h_m}{2} = \frac{1}{40} \Rightarrow h_m = \frac{2}{40} = 0.05 \text{ m}$$

Example

A spillway model is to be built to a geometrically similar scale of $\frac{1}{40}$ across a flume, of 50 cm width. The prototype is 20 m high and maximum head on it is expected to be 2 m. (i) What height of model and what head on the model should be used ? (ii) If the flow over the model at a particular head is 10 litres/s, what flow per metre length of the prototype is expected ? (iii) If the negative pressure in the model is 150 mm, what is the negative pressure in the prototype ? Is it practicable ?

[Ans. (i) 0.5 m, 0.05 m, (ii) 5059.64 litres/s (iii) 6 m. Yes]

$$\frac{Q_m}{Q_p} = \frac{A_m \times V_m}{A_p \times V_p} = \frac{L_m^2}{L_p^2} \times \frac{V_m}{V_p} = \left(\frac{L_m}{L_p}\right)^{2.5}$$

$$\frac{V_p}{\sqrt{L_p g_p}} = \frac{V_m}{\sqrt{L_m g_m}} \Rightarrow \frac{V_m}{L_p} = \sqrt{\frac{L_m}{L_p}}$$

Example

A spillway model is to be built to a geometrically similar scale of $\frac{1}{40}$ across a flume, of 50 cm width. The prototype is 20 m high and maximum head on it is expected to be 2 m. (i) What height of model and what head on the model should be used ? (ii) If the flow over the model at a particular head is 10 litres/s, what flow per metre length of the prototype is expected ? (iii) If the negative pressure in the model is 150 mm, what is the negative pressure in the prototype ? Is it practicable ?

[Ans. (i) 0.5 m, 0.05 m, (ii) 5059.64 litres/s (iii) 6 m. Yes]

$$\frac{L_m}{L_p} = \frac{1}{40} = \frac{0.15}{h_p} \Rightarrow h_p = 40 \times 0.15 = 6 \text{ m}$$



THANK YOU